# Foundations

of data communication networks

layout: two-cols-title columns: is-4

*Doing that thing Computer Scientists do...*

layout: quote color: sky-light quotesize: text-m authorsize: text-s author: 'Talking Heads'

"*How did I get here?*"

layout: two-cols-title columns: is-4

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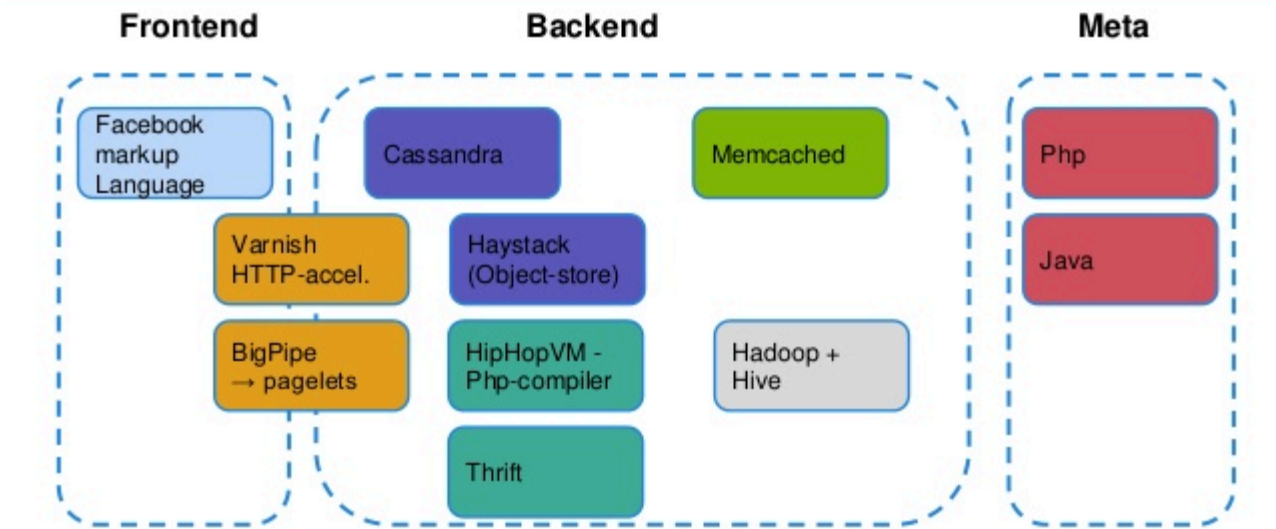
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layout: center

# Facebook




Source: Gerald Maduabuchi(Quora), Phillip Webber

layout: center

# Tech Stacks

In groups of 2-3, Google the "tech stack" for a website or web based application. In your investigation, answer these two questions:

**Question**

## What technologies do they use?

**Question**

## Which are *network* technologies?

05:00

▶ Start

layout: default

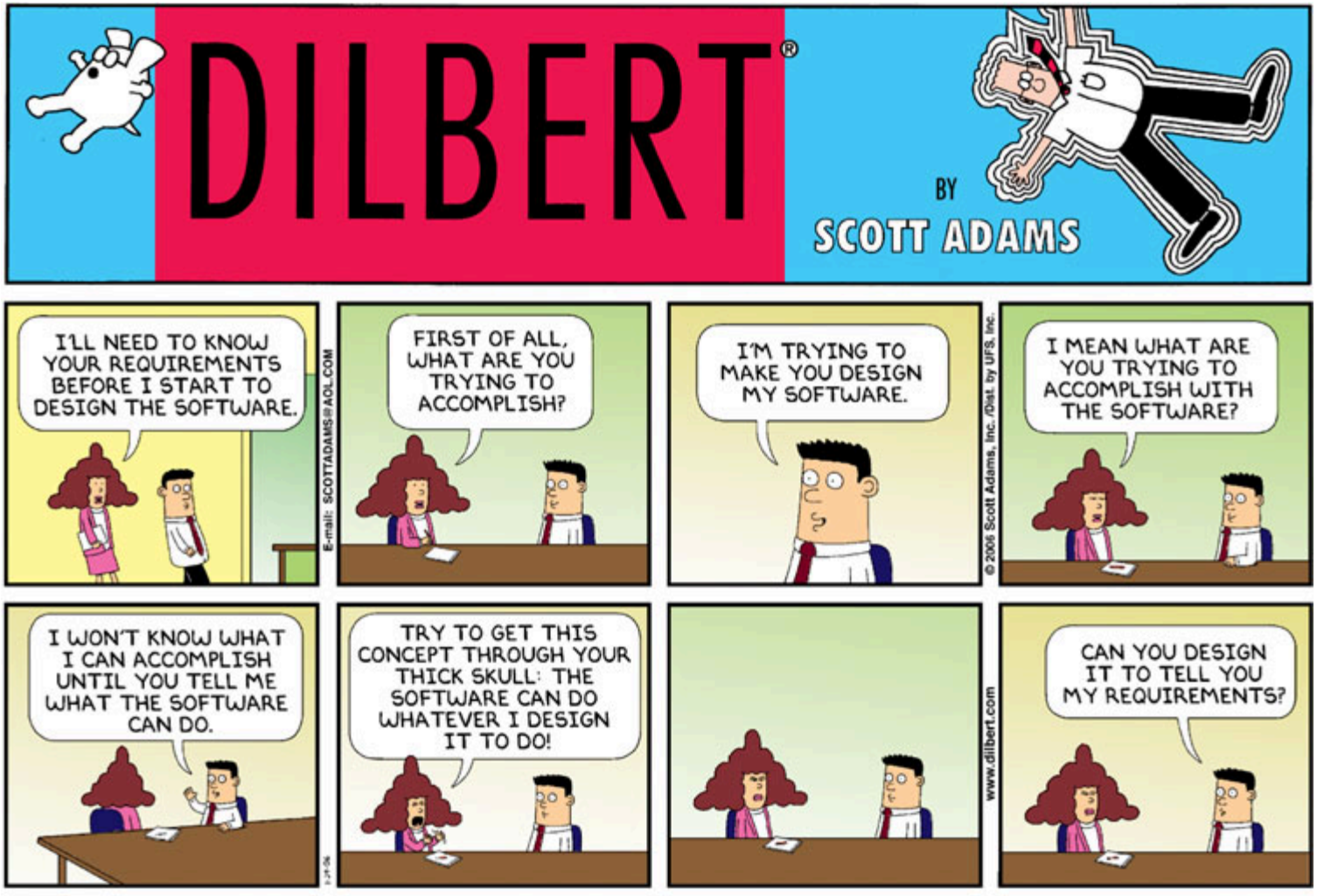
# Tech Stacks

- What technologies did you find?
- Which ones "cross the network"?
- Were there any you found as *standards* (e.g. `http`)
- Were there any that were *unique* to that website or application?
- What else did you find?

layout: two-cols-title columns: is-4

# Let's start looking at *requirements* first

layout: center title: dilbert requirements


DILBERT®
BY
SCOTT ADAMS
I'LL NEED TO KNOW YOUR REQUIREMENTS BEFORE I START TO DESIGN THE SOFTWARE.
FIRST OF ALL, WHAT ARE YOU TRYING TO ACCOMPLISH?
I'M TRYING TO MAKE YOU DESIGN MY SOFTWARE.
I MEAN WHAT ARE YOU TRYING TO ACCOMPLISH WITH THE SOFTWARE?
I WON'T KNOW WHAT I CAN ACCOMPLISH UNTIL YOU TELL ME WHAT THE SOFTWARE CAN DO.
TRY TO GET THIS CONCEPT THROUGH YOUR THICK SKULL: THE SOFTWARE CAN DO WHATEVER I DESIGN IT TO DO!
CAN YOU DESIGN IT TO TELL YOU MY REQUIREMENTS?
E-mail: SCOTTADAMS@AOL.COM
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layout: two-cols-title title: requirements

layout: two-cols columns: is-7 title: common requirements

layout: two-cols-title columns: is-4 title: scalable connectivity

layout: two-cols-title columns: is-4 title: cost effective

layout: two-cols columns: is-8

layout: two-cols columns: is-8 title: sharing the wire - round 1

layout: two-cols columns: is-8 title: sharing the wire - round 2

layout: two-cols columns: is-8 title: sharing the wire - round 3

title: sharing the wire - discussion

# Sharing the Wire – Discussion

- **Round 1** - Frequency Division
  - Each sender has $1/3$ the total **bandwidth** and sends $1/3$ of a packet at a time.
  - Wasted lane when sender has no more data to send. **not efficient**
- **Round 2** - Time Division
  - Each sender has full bandwidth for $1/3$ of the time.
  - Wasted time slots when sender has no more data to send. **not efficient**
- **Round 3** - Packet Switching
  - No wasted lane or time slots
  - Senders collide when sending.
  - Receiver may need to put packets back in order.

layout: two-cols-title columns: is-4 title: frequency division

layout: two-cols-title columns: is-4 title: time division

layout: two-cols-title columns: is-4 title: time division compact

layout: two-cols-title columns: is-4 title: packet switching

layout: two-cols-title columns: is-4 title: multiplexing summary

layout: two-cols-title columns: is-4 title: support common services

layout: two-cols-title columns: is-4 title: reliable and manageable

layout: two-cols title: common requirements review

layout: two-cols title: user requirements

layout: two-cols title: developer requirements

layout: two-cols title: provider requirements

layout: center title: missing requirements

# Requirements

Take a couple of minutes to think about and write down

**Question**

## What requirements are we missing?

02:30

▶ Start

layout: two-cols columns: is-7 title: architecture

layout: two-cols-title columns: is-4 title: architecture layers

layout: two-cols-title columns: is-4 title: architecture protocols

layout: two-cols-title columns: is-4 title: architecture encapsulation

layout: two-cols-title columns: is-4 title: architecture encapsulation

layout: two-cols-title columns: is-4 title: architecture encapsulation multiplex

layout: two-cols-title columns: is-4 title: architecture encapsulation demultiplex

layout: two-cols-title columns: is-4 title: architecture ISO/OSI model

layout: two-cols-title columns: is-4 title: architecture ISO/OSI model

layout: two-cols-title columns: is-4 title: architecture internet model

layout: two-cols-title columns: is-4 title: architecture comparison

layout: two-cols columns: is-7 title: performance

layout: two-cols-title columns: is-4 title: performance bandwitdh

layout: two-cols-title columns: is-4 title: performance bandwitdh example

layout: two-cols-title columns: is-4 title: performance latency

layout: two-cols-title columns: is-4 title: performance latency speed of light

layout: two-cols-title columns: is-4 title: performance latency typical

layout: two-cols-title columns: is-4 title: performance final equation

layout: two-cols-title columns: is-4 title: performance cards setup

layout: two-cols-title columns: is-4 title: performance cards round 1

layout: two-cols-title columns: is-4 title: performance cards round 1 answer

layout: two-cols-title columns: is-4 title: performance cards round 1 summary

layout: two-cols-title columns: is-4 title: performance cards setup latency

layout: two-cols-title columns: is-4 title: performance cards round 1

layout: two-cols-title columns: is-4 title: performance cards round 2 answer

layout: two-cols-title columns: is-4 title: performance break activity

layout: two-cols-title columns: is-4 title: performance problem 1

layout: two-cols-title columns: is-4 title: performance bandwidth x latency

layout: two-cols-title columns: is-4 title: performance bandwidth x latency question

layout: two-cols-title columns: is-4 title: performance exercise intro

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layout: two-cols-title columns: is-4 title: performance exercise a

layout: two-cols-title columns: is-4 title: performance exercise b

layout: two-cols-title columns: is-4 title: performance exercise c

layout: two-cols-title columns: is-4 title: performance exercise d

layout: two-cols-title columns: is-4 title: performance exercise 2

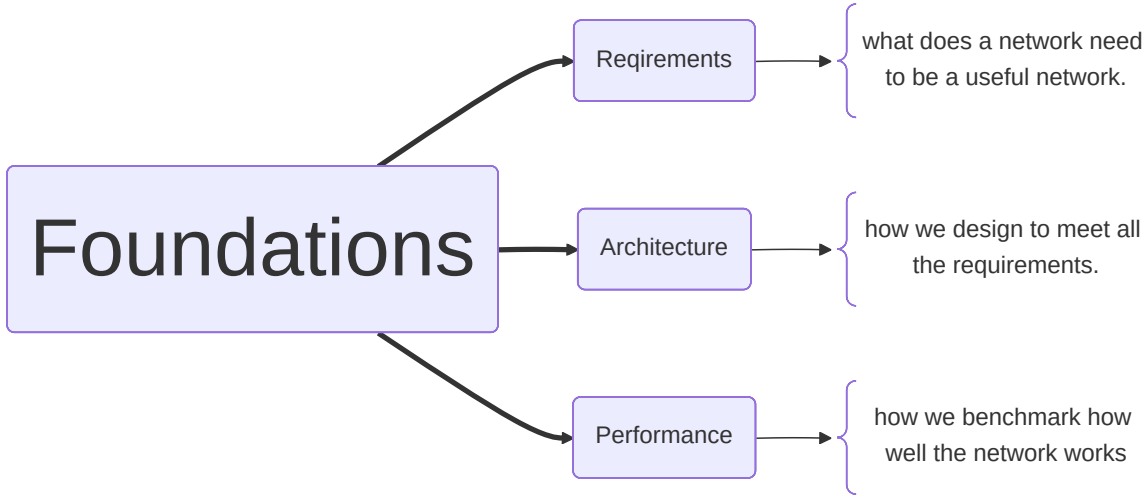
layout: two-cols-title columns: is-4 title: performance exercise 2 a

layout: two-cols-title columns: is-4 title: performance exercise 2 b

layout: two-cols-title columns: is-4 title: performance exercise 2 c

layout: two-cols-title columns: is-4 title: performance exercise 2 d

layout: center


Foundations
Reqirements
what does a network need to be a useful network.
Architecture
how we design to meet all the requirements.
Performance
how we benchmark how well the network works